MILITARY SPECIFICATION

VALVES, PNEUMATIC 4-WAY, 2-POSITION,

MANUALLY OPERATED

1. SCOPE

1.1 This specification covers actuation valves with 4500 pounds per square inch gage (psig) ratings used for remote pneumatic operation of submarine EMBT blow valves.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-P-40 - Packaging and Packing of Hand Tools.

MILITARY
MIL-P-116 - Preservation, Methods of.
MIL-R-196 - Repair Parts For Internal Combustion Engines, Packaging of.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-D-1000/2 - Drawings, Engineering and Associated List.
MIL-M-15071 - Manuals, Technical: Equipments and Systems Content Requirements For.
MIL-B-23921 - Bronze, Nickel Aluminum Castings for Seawater Service.
MIL-B-24059 - Bronze-Nickel Aluminum; Rod, Flat Products with Finished Edges, Shapes and Forgings.

STANDARDS

MILITARY
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

DRAWINGS

MILITARY
810-1385884 - Unions, Fittings and Adapters, Butt and Socket Welding 6000 PSI WOG and Oxygen (IPS).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 222 South Riverside Plaza, Room 1106, Chicago, Illinois 60606.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 471-68 - Change in Properties of Elastomeric Vulcanizates Resulting From Immersion in Liquids.
D 1414-68 - Rubber O-Rings.
D 2240-68 - Indentation Hardness of Rubber and Plastic by Means of a Durometer.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. The valves furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 Design concept. All internal trim shall be readily replaceable with the valve in the line. The design shall permit loss, displacement, or severe damage to any one or more of the non-metallic internal sliding seals or seats without jeopardizing the valve's functioning and ability to meet the criteria of 3.15. Emphasis shall be placed on ruggedness, reliability and simplicity of design.

3.3 Actuation. The main spool or poppet shall be directly connected to the actuating lever so that intermediate spring forces are not required for positioning or affecting a seal. The operation of the valve shall not be dependent on any critical adjustments. Valve shall be actuated by means of a short throw lever. Total travel of lever between the two positions shall not exceed 90 degrees. A detent construction shall be provided to lock valve in each position and positively prevent an inadvertent shift of position.

3.4 Materials of construction. Materials shall be as specified in table I. All materials shall be selected to prevent corrosion, galling, seizing or excessive wear.

Table I - List of materials.

Name of parts	Material
Pressure containing envelope (body, top and bottom covers, and so forth)	70-30 nickel-copper 70-30 copper-nickel CRES (300 series) Aluminum Bronze Cast: MIL-B-23921 Forged: MIL-B-24059
Internal trim	70-30 nickel-copper CRES (300 series) Stellite 17-4PH
External bolting	CRES (300 series) Cadmium plated steel
Lever, lever bracket and other parts external to pressure containing envelope	CRES (300 series) Aluminum Brass Aluminum-bronze Bronze 70-30 nickel-copper 70-30 copper-nickel
Static and dynamic seals	Materials suitable for this application and meeting all performances and compatibility criteria specified herein.

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3.5 Design and test pressures. The design and test pressures of the pressure containing envelope shall be as follows:

Nominal rating Psig	Proof pressure Psig	Burst pressure Psig
4500	6750	13,500

3.6 Ports. Valves shall be equipped with 1/4 inch iron pipe size (ips) 6000 psig union connections in accordance with Drawing 810-1385884. Only the pertinent dimensions of this drawing apply. Union thread pieces may be forged integral with the valve body or secured to the body with a straight thread and O-ring seal. Where a straight thread and O-ring seal is used, the O-ring shall be inboard of the threads.

3.7 Capacity. Valves shall have an Equivalent Sharp Edge Orifice Diameter (ESEOD) through each flow path (pressure port to cylinder 1 port, pressure port to cylinder 2 port, cylinder 1 port to vent port, cylinder 2 port to vent port) of at least 0.2 inches.

3.8 Interflow. The valve design shall prevent interflow. At no position during spool shift shall there be communication between pressure and vent.

3.9 Mechanical shock and vibration. Valves shall be designed to meet the mechanical shock requirements of MIL-S-901 and the vibration requirements of MIL-STD-167. There shall be no damage, shift in position, or degradation of operational capacity to the valve when subjected to the above conditions. Momentary internal leakage is acceptable, providing the failure mode criteria of 3.15 is not exceeded during the transient and that the valve returns immediately to full performance.

3.10 Actuation force. The tangential force required to shift positions of the valve shall not exceed 15 pounds under any operating condition.

3.11 Stem guiding. The spool shall be positively guided to prevent cocking or binding under any tolerance stack-up condition.

3.12 Threads. All threads shall conform to Handbook H28. Pipe threads shall not be used. Accidental loosening of all threaded parts shall be prevented by safety wiring on a prevailing-torque locking feature.

3.13 Interchangeability. The valve shall incorporate complete parts interchangeability. Each part shall be identified by part number, and shall be replaceable from stock or the manufacturer on a non-selective and random basis. All parts having the same manufacturer's part number shall be directly interchangeable with each other with respect to installation (physical) and performance (function). Physical interchangeability shall be in accordance with the following definition: Interchangeable assemblies, components, and parts are those which are capable of being readily installed, removed, or replaced without alteration, misalignment, or damage to parts being installed or to adjoining parts. No fabrication operations such as cutting, filing, drilling, reaming, hammering, bending, prying, or forcing shall be required. This is not intended to preclude the use of special tools, fixtures, and other shop aids during original assembly of the parts into the article. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)).

3.14 Seat tightness. All seals (static and dynamic) shall provide zero internal and external leakage (defined as no visible signs of leakage over a 5-minute period using a bubble-fluid method or submerging the valve in water).

3.15 Failure-mode. The valve shall be capable of failure-mode performance (defined as providing not less than 3000 psig at the on port and not more than 100 psig at the off port with a 4500 psig air supply) with loss, displacement, or severe damage to any one or more of its non-metallic sliding seals or seating elements or any damage or wear caused by line media contamination.

3.16 Marking.

3.16.1 Body markings. Valve bodies shall be marked to show the cylinder 1, cylinder 2, pressure and vent or return ports. In addition markings shall be provided to identify the different lever positions.

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3.16.2 Identification plates. An identification plate of corrosion resisting steel, brass, or anodized aluminum shall be securely attached to the valve and shall list the following data or a space therefor:

(a) Manufacturer's name or trademark.
(b) MIL-V-24232
(c) Valve serial number.
(d) Manufacturer's drawing number.
(e) Manufacturer's model and part number.
(f) Applicable manual number.
(g) Space for 9-digit CID number.

3.17 Drawings.

3.17.1 Preliminary drawings. Preliminary drawings which are sufficient to permit evaluation of the design and of materials shall be submitted to the procuring activity. Drawings shall show the following:

(a) A sectional assembly of the valve.
(b) Finish and hardness of the dynamic guiding surfaces.
(c) Clearances (maximum and minimum) of the dynamic guiding surfaces.
(d) Concentricity and parallel requirements for the dynamic guiding surfaces in the assembled valve.
(e) End connection detail.
(f) Bill of material listing specification, grade, condition and any other data required to fully identify the properties of the materials proposed.
(g) ESEOD for each flow path.
(h) Installation dimensions, end connection detail, and clearance dimensions required for disassembly.
(i) Recommended assembly, torques, or other equivalent procedures, for making up all joints and threaded assemblies.

3.17.2 Final drawings. Final drawings and certification data sheets shall be submitted to the procuring activity within 60 days after date of contract. These drawings shall be in accordance with types II and III of MIL-D-1000/2 except for extent of detail. Only the information required in 3.17.1 need be furnished for the type II drawings. The following data, in addition to that required in MIL-D-1000/2, shall be furnished for the type III drawings:

(a) Applicable assembly drawing number(s).
(b) Tabulated data from Infra Red (IR) test analysis on all elastomeric materials in valve (see 4.2.13).

3.17.3 Limited rights legend. When the Government has only limited rights in the data shown on the drawings, as determined by the contractural provisions regarding rights in technical data, the drawings furnished may be marked with the following restrictive legend:

"Furnished under United States Government Contract No. _____. Shall not be either released outside the Government, or used, duplicated, or disclosed in whole or in part for manufacture or procurement, without the written permission of _____, except for: (a) emergency repairs or overhaul work by or for the Government, where the item or process concerned is not otherwise reasonably available to enable timely performance of the work; or (b) release to a foreign government, as the interests of the United States may require; provided that in either case the release, use duplication or disclosure hereof shall be subject to the foregoing limitations. This legend shall be marked on any reproduction hereof in whole or in part."

3.18 Manuals. Manuals shall be furnished in accordance with type I of MIL-M-15071. The quantity and distribution of manuals shall be as specified (see 6.2). The following, in addition to that required for type I of MIL-M-15071, shall be included as part of the manual contents:

(a) The engineering drawings for the valve (including certification data sheet). These drawings shall be supplemented by additional illustrations where necessary to adequately illustrate operation and maintenance. These additional illustrations may consist of blowouts, partial or full sections, etc., and may eliminate extraneous lines and details to clarify the interaction of parts.

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(b) Table listing wrench sizes and assembly torques (or other equivalent procedures) for making up all joints and threaded assemblies.
(c) Instructions to permit overhaul by shipyard or other repair facilities. These should include procedures for checking all critical dimensions subject to wear or change and the acceptable dimensional limits, surface finish condition, etc. Also, the appropriate procedure (that is, part replacement, correction at repair facility, or repair at manufacturer's facility) which should be followed to correct each case of damage or wear.
(d) Detailed disassembly and reassembly procedures. In addition to providing procedures for the complete disassembly and reassembly of the equipment, maintenance and troubleshooting sections shall contain, or refer to, only the limited disassembly and reassembly required to accomplish each particular operation. This is intended to reduce the possibility of unnecessary disassembly and unnecessary disturbance of adjustments when performing specific or limited maintenance or troubleshooting operations.
(e) Adjustment procedures.

3.19 Installation pattern. The valve configuration shall be in accordance with pattern A or B as specified (see figures 1 and 2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests.[1/] Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the examination and tests specified in 4.2.1.

4.2.1 Samples for qualification examination and tests. One actuation valve shall be submitted for the examination and tests outlined in table II and as specified in 4.2.2 through 4.2.13. Acceptance criteria shall be as given in table II. Detail engineering drawings shall be submitted with the valve(s) to be tested, in addition to 100 percent repair parts.

Table II - Qualification examination and test outline.

Examination and tests	Test conditions	Purpose of examination and tests	Acceptance criteria
Dimensional examination (see 4.2.2)	- - - -	To verify dimensional conformance of test valves to detailed engineering drawings	Complete dimensional conformance
Proof test (see 4.2.3)	6750 psig water applied to pressure containing envelope	To test strength and soundness of pressure containing envelope	No external leakage or structural failure
Leakage test (see 4.2.4)	4500 psig air to pressure port for each valve position	To test integrity of internal seals	No visible leakage over a 5 minute period with valve submerged in water
Failure mode test (see 4.2.5)	4500 psig air to pressure port with worst combination of seal damage/removal. Check at both positions.	To test for: (a) Maximum pressure build-up at actuation "off" port upon loss/damage of seat seal	Pressure build-up at "off" port shall not exceed 100 psig
		(b) Minimum pressure build-up up at actuation"on" port upon loss/damage of seat seal	Pressure build-up at "on" port shall be 3000 psig or greater

1/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3 and 6.4).

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Table II - Qualification examination and test outline (Cont'd).

Examination and tests	Test conditions	Purpose of examination and tests	Acceptance criteria
Flow capacity (see 4.2.6)	Air	To determine ESEOD of flow paths through the valve	ESEOD 0.2 inches minimum
Endurance test (see 4.2.7)	4500 psig air, 1000 cycles	To determine endurance of internals, particularly the dynamic seals	Proper functioning and zero leakage during and after 1,000 cycles
Contamination (see 4.2.8)	4500 psig air, 500 cycles Contaminants: siliceous dust and red iron oxide Concentration- 0.001 percent	To determine susceptibility of valve to damage and malfunction from system contaminants	Valve to remain operable; zero leakage; meet failure mode requirements
H.I. shock test (see 4.2.10)	Grade A, class I of MIL-S-901	To determine resistance of valve to high-impact mechanical shock	Zero leakage and no position change or stru-- tural damage
Vibration test (see 4.2.11)	Type I of MIL-STD-167; pressurized with 4500 psig air	To determine resistance of valve to environmental vibration conditions	No resonant frequency 0-33 cps Zero leakage and no position change or structural damage
Seal materials tests (see 4.2.12)	(See 4.2.12)	To verify that seal material used will provide satisfactory long term service	(see 4.2.12)
Infra Red test (see 4.2.13)	- - - -	To establish make-up of seals used in qualification tests	- - - -

4.2.2 Examination prior to testing. Upon receipt of the sample submitted for qualification testing (see 4.2.1), the sample valve shall be disassembled and examined to verify complete dimensional conformance to the detailed engineering drawings. Particular emphasis shall be placed on dimensions, finishes and concentricity of the guiding and seating surfaces.

4.2.2.1 Upon satisfactory completion of the examination specified in 4.2.2, the valve shall be tested as specified in 4.2.3 through 4.2.13.

4.2.3 Proof test. With the cylinder ports (C1 and C2) cross connected and the vent port capped, 6750 psig water shall be applied to the pressure port for 2 minutes, depressurized and 6750 psig water shall again be applied for 2 minutes.

4.2.4 Leakage test. With both cylinder ports (C1 and C2) capped, 4500 psig air shall be applied to the pressure port and the vent port shall be checked for leakage over a 5 minute period. This test shall be conducted with the lever detented in both the C1 and the C2 position.

4.2.5 Failure mode test. The non-metallic seal between the pressure port and C1 port shall be removed or damaged, as approved. The C1 and C2 port shall be blocked with gages and the lever shall be detented in the C2 position. 4500 psig air shall be applied to the pressure port and the pressures shall be recorded at the C1 and C2 ports. Test shall be repeated removing the non-metallic seal which seals between the pressure port and the C2 port and detenting the lever in the C1 position.

4.2.6 Flow capacity. The ESEOD shall be determined, using air of the following flow paths:

(a) Pressure port to C1 port.
(b) Pressure port to C2 port.
(c) C1 port to vent.
(d) C2 port to vent.

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4.2.7 Endurance test. Each cylinder port (C1 and C2) shall be connected, by a full unrestricted 1/4 inch line, to separate 0.02 cubic foot dead end volumes. 4500 psig air shall be applied to the pressure port and the valve shall be cycled 1000 times. Each cycle shall consist of the change in lever position from C1 to C2 and back to C1. The valve shall remain in position-under pressure-at least 5 minutes between each 1/2 cycle, in order to permit the internal seals to take a set. After each 250 cycles, the valve shall be allowed to remain under pressure in the C1 position for 1 hour, and then reshuttled back to the C2 position, and then back to the C1 position. The valve shall be monitored for leakage and actuation force during the entire test, and values shall be recorded at least each 25 cycles.

4.2.8 Contamination cycle test. Each cylinder port (C1 and C2) shall be connected to separate 0.02 cubic foot dead end volumes (see 4.2.7). 4500 psig air shall be applied to the pressure port and the valve shall be cycled 500 times. Each cycle shall consist of the change in lever position from C1 to C2 and back to C1. The valve shall remain in position at least 5 minutes between each 1/2 cycle. The valve shall be monitored for leakage and actuation force during the entire test, and valves shall be recorded at least each 25 cycles. During this test, contaminants shall be added to the air entering the valve. Composition and concentration of these contaminants shall be as follows:

Cycle	Contaminant	Minimum contaminant concentration
0-250	Siliceous dust	0.00001 #/# air
250-500	Red iron oxide	0.00001 #/# air

At completion of each 250 cycles, tests of 4.2.4 and 4.2.5 and examination of 4.2.9 shall be repeated.

4.2.9 Visual and dimensional examination. Valve shall be disassembled and visually and dimensionally examined for damage, wear and operation.

4.2.10 High-impact shock test. The valve shall be subjected to and meet the high-impact shock requirements for grade A, class I of MIL-S-901 with C1 and C2 ports capped and 4500 psig applied to the pressure port. The valve shall be tested in both the C1 and C2 positions.

4.2.11 Vibration test. The valve shall be vibration tested in accordance with type I of MIL-STD-167 with the C1 and C2 ports capped and 4500 psig applied to the pressure port. The valve shall be checked for leakage in both the C1 and C2 positions during the endurance portion of the test.

4.2.12 Seal material tests. All elastomers used in the valve shall successfully pass the tests specified in 4.2.12.1 through 4.2.12.7.

4.2.12.1 Initial properties. The ambient temperature during the determination of properties shall be in range of 80° ± 9°F.

4.2.12.1.1 Tensile strength and ultimate elongation. These properties shall be determined in accordance with ASTM D1414-68. The test specimens shall be six ARP568-229 O-rings. The average values obtained shall be reported.

4.2.12.1.2 Hardness. This property shall be determined on a 1/4 inch thick specimen in accordance with ASTM D2240-68 using a type A durometer equipped with a 1 kilogram weight centered on the axis of the indentor. The durometer reading shall be taken 15 seconds after the presser foot is in firm contact with the specimen. The test specimens shall be two discs or slabs, 1/4 inch thick. The average value shall be reported to the nearest point.

4.2.12.1.3 Tension set. This property shall be determined in accordance with ASTM D1414-68. The test specimens shall be three ARP568-229 O-rings. The mean value shall be reported to the nearest 0.5 percent.

4.2.12.2 Hydrolytic stability test. Hydrolytic stability test shall be conducted as follows: Place 250 milliliters of tap water in a liter desiccator. Mix in the proper amount of potassium sulfate to assure 95 ± 5 percent relative humidity in the desiccator. Place a rack in the desiccator above the surface of the liquid and suspend six slabs or discs so that they do not contact the liquid. Place covered desiccator in a circulating air oven set at 200° ± 5°F. for 30 days. At the end of this 30 day test period, remove samples. There shall be no evidence of fungus growth. Dry for 2 hours at room temperature. Shore A hardness shall be measured and shall not show more than a 50 percent reduction.

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4.2.12.3 Properties after aging in high pressure air. The test specimens shall be aged in a high-pressure container for 168 + 1/2 hours at 180° + 2°F. under 6000 + 50 psig air pressure. At the end of the 168 hours, the air pressure shall be released at a rate not to exceed 200 psig per hour. The 180°F. temperature shall be maintained until the pressure has been reduced to 0 psig. Tensile strength and ultimate elongation retention shall be not less than 90 percent.

4.2.12.3.1 Tensile strength and ultimate elongation. During aging the O-rings shall be suspended in the container without touching each other or the sides of the container. At the termination of the aging interval, the O-rings shall be removed from the container, cooled to room termperature on a flat surface and allowed to rest not less than 16 hours nor more than 96 hours before testing. Six ARP568-229 O-rings shall be aged and tested. The procedure for measuring tensile strength and ultimate elongation after aging is specified in 4.2.12.1.1. The average test values shall be used for calculating the retention of properties after aging.

4.2.12.3.2 Hardness. During aging the discs or slabs shall be supported on edge by the bottom of the container in such a manner that a minimum area of the rubber contacts the container. At the termination of the aging interval, the discs or slabs shall be removed from the container, cooled to room termperature on a flat surface and allowed to rest not less than 16 hours nor more than 96 hours before testing. Two 1/4 inch thick discs or slabs shall be aged and tested. The procedure for measuring hardness is specified in 4.2.12.1.2. The average test value shall be recorded. Change in hardness shall not exceed 10 durometer points.

4.2.12.3.3 Compression set. One segment (about 120 degrees) from each of three ARP568-331 O-rings shall be tested. The compression plates, spacers, methods of measuring the specimens and assembling the apparatus, and the treatment of the specimens after release from the plates shall be those specified in ASTM D1414-68. The O-ring segments shall be released from the warm compression plates (180°F.) within 10 minutes after the high-pressure container is opened. The mean value for compression set shall be reported. Compression set shall not exceed 30 percent.

4.2.12.4 Increase in inside diameter after exposure to high pressure air.

4.2.12.4.1 The inside diameter of each of three ARP568-229 O-rings shall be determined to the nearest 0.003 inch using a metal stepped cone gage machined in seven 0.003 inch steps, each 0.25 inch wide. The seven steps shall range from 2.349 inches to 2.370 inches in diameter, to cover the allowed tolerance for the inside diameter of ARP568-229 O-rings.

4.2.12.4.2 The O-ring specimens shall be placed in a high-pressure container and subjected to 6000 + 50 psig air pressure for 96 + 1/2 hours at 80° + 9°F. At the end of this period, the air pressure in the container shall be lowered to 0 psig in less than 15 seconds. The specimens shall then be removed from the container and their inside diameters measured by means of a stepped cone gage exactly 5 minutes + 15 seconds after the pressure in the container has reached 0 psig. The stepped cone gage used shall range in diameter from 2.443 to 2.464 inches in seven 0.003-inch steps, each 0.25 inch wide, to cover the maximum allowable increases of 0.094 inch for O-rings within the allowed tolerance for the inside diameter of the ARP568-229 O-rings. The increases in the inside diameters of the three O-rings shall be computed and the mean value found shall be reported. Increase inside diameter shall not exceed 0.094 inches.

4.2.12.5 Properties after immersion in water. The test specimens shall be immersed in distilled water at 180° + 2°F. for 168 + 1 hours. The conditions of immersion shall be in accordance with ASTM D1414-68 and ASTM D471-68. The test specimens shall be completely surrounded by the fluid during the immersion. The fluid volume shall be not less than 12 times the total volume of the specimens. After the immersion period and before testing, all specimens shall be conditioned in fresh distilled water at 80° + 9°F. for 30 to 90 minutes.

4.2.12.5.1 Volume change. This property shall be determined using three ARP568-229 O-rings in accordance with ASTM D1414-68. The mean value shall be reported. Volume change shall not exceed 40 percent.

4.2.12.5.2 Tensile strength and ultimate elongation. These properties shall be determined in accordance with ASTM D1414-68 and ASTM D471-68 by the "immediate deteriorated properties" method. Tensile strength shall be based on the final cross sections of the specimens. The final cross sections shall be the original cross sections corrected for the change in cross section due to immersion. The latter value shall be calculated from the volume change found in 4.2.12.5.1. The test specimens shall consist of six ARP568-229

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O-rings. The average values found for tensile strength and ultimate elongation shall be used for calculating the retention of original properties. Tensile strength retention shall be not less than 45 percent and ultimate elongation retention shall be not less than 65 percent.

4.2.12.5.3 Hardness. Two 1/4 inch thick discs or slabs shall be immersed and then tested as specified in 4.2.12.1.2. The specimens shall be tested within 3 minutes after removal from the fluid. The average value found shall be used for determining the change in hardness. Change in hardness shall not exceed 15 durometer points.

4.2.12.6 Properties after immersion in ASTM No. 1 oil.[2/] The test specimens shall be immersed in ASTM No. 1 oil at 180° + 2°F. for 168 + 1 hours. The conditions of immersion shall be in accordance with ASTM D1414-68 and ASTM D471-68. The test specimens shall be completely surrounded by the fluid during immersion. The fluid volume shall be not less than 12 times the total volume of the specimens. After the immersion period and before testing, all specimen shall be conditioned in fresh ASTM No. 1 oil at 80° ± 9°F. for 30 to 90 minutes.

4.2.12.6.1 Volume change. This property shall be determined by the procedure specified in 4.2.12.5.1. Volume change shall not exceed 5 percent.

4.2.12.6.2 Tensile strength and ultimate elongation. These properties shall be determined by the procedure specified in 4.2.12.5.2. Tensile strength and ultimate elongation retention shall be not less than 75 percent.

4.2.12.6.3 Hardness. This property shall be determined by the procedure specified in 4.2.12.5.3. Change in hardness shall not exceed 5 durometer points.

4.2.12.7 Properties after immersion in freon PCA. The test specimens shall be immersed in Freon PCA (purified Freon TF) at 80° + 9°F. for 168 + 1 hours. The conditions of immersion shall be in accordance with ASTM D1414-68 and ASTM D471-68. The test specimens shall be completely surrounded by the fluid during immersion. The fluid volume shall be not less than 12 times the total volume of the specimens.

4.2.12.7.1 Volume change. This property shall be determined by the procedure specified in 4.2.12.5.1. Volume change shall not exceed 10 percent.

4.2.12.7.2 Tensile strength and ultimate elongation. These properties shall be determined by the procedure specified in 4.2.12.5.2. Tensile strength and ultimate elongation retention shall be not less than 25 percent.

4.2.12.7.3 Hardness. This property shall be determined by the procedure specified in 4.2.12.5.3. Change in hardness shall not exceed 5 durometer points.

4.2.13 Infra Red (IR) test. Samples of each elastomeric material used in the qualification valve, and from the same batch as those tested in 4.2.12, shall undergo IR analysis to determine their make up. The results of the IR analysis shall be tabulated on the approved drawing for reference against which future batch analysis can be compared.

4.3 Sampling for quality conformance inspection.

4.3.1 Lot. All valves of the same size offered for delivery at one time, shall be considered a lot for the purpose of sampling.

4.3.2 Sampling for visual and dimensional examination. A random sample of valves shall be selected from each lot in accordance with table III and shall be examined as specified in 4.4.1.1 and 4.4.1.2. Failure of any valve in the sample to pass the examination specified in 4.4.1.1 and 4.4.1.2 shall be cause for rejection of the lot.

Table III - Sampling for visual and dimensional examination.

Lot size	Sample quantity
2 to 25	1
26 to 65	2
66 to 180	3
Over 180	4

[2/] ASTM reference oil no. 1 as specified in ASTM D471-68.

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4.4 Quality conformance inspection.

4.4.1 Examination.

4.4.1.1 Visual examination. A visual examination shall be made of the sample valves selected in accordance with 4.3.2 to verify conformance to the requirements of this specification.

4.4.1.2 Dimensional examination. A dimensional examination shall be made on the sample valves selected in accordance with 4.3.2 to verify conformance of the guiding and sealing diameter to the requirements of the guiding and sealing diameters to the requirements of the accepted master drawings.

4.4.2 Tests.

4.4.2.1 Proof test. Each valve shall be pressurized hydrostatically to 6750 psig for 2 minutes, depressurized and again pressurized to 6750 psig for 2 minutes with the cylinder ports cross connected and the vent port capped. There shall be no evidence of external leakage or structural failure during this test.

4.4.2.2 Leakage and functional test. Each valve shall be connected to a 4500 psig air source and cycled 10 times using the ullage at the cylinder ports defined in 4.2.7. The valve shall then be checked for leakage at both the C1 and C2 positions. External leakage shall also be checked. There shall be no visible leakage over a 2 minute period in each position. The required actuation force during this test shall be noted and shall not exceed the value specified in 3.9.

4.4.2.3 Material conformance test. One complete set of elastomeric parts from each batch or production lot shall undergo IR analysis and results compared to the IR analysis established during qualification. There shall be no deviation greater than plus or minus 5 percent in the IR comparison readings.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 Preservation-packaging, packing and marking. Preservation-packaging shall be level A or C, packing level A, B or C and marking in accordance with MIL-V-3 as specified (see 6.2).

5.1.1 Repair parts and tools. Repair parts and tools shall be preserved-packaged, packed and marked in accordance with MIL-R-196 and PPP-P-40 in the levels specified as follows:

	Preservation-package	Packing
Onboard (accompanying)	A	C
Stock	A	B

5.2 Cushioning or dunnage. Cushioning or dunnage material shall be in accordance with MIL-STD-1186 except use of polystyrene "loose-fill" material, unless approved by the procuring activity for packaging and packing applications is prohibited. When approved, containers (interior and exterior) shall be marked or labelled as follows:

"CAUTION

Contents cushioned with polystyrene "loose-fill" material not to be taken onboard ship.
Remove and discard cushioning before shipboard storage.
If required, recushion with cellulosic material, bound fiber, fiberboard transparent flexible cellular material, etc."

6. NOTES

6.1 Intended use. Valves specified herein are intended for use in submarine EMBT actuation lines.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Manuals (quantity and distribution) (see 3.18).
(c) Preservation, packaging, packing and marking requirements, if other than specified in 5.1 (see 5.2).

6.2.1 Bid data. Bid data should include the preliminary drawings specified in 3.17.1.

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL 24232 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.4).

6.4 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE WERE MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
Navy - SH
(Project 4820-N238)

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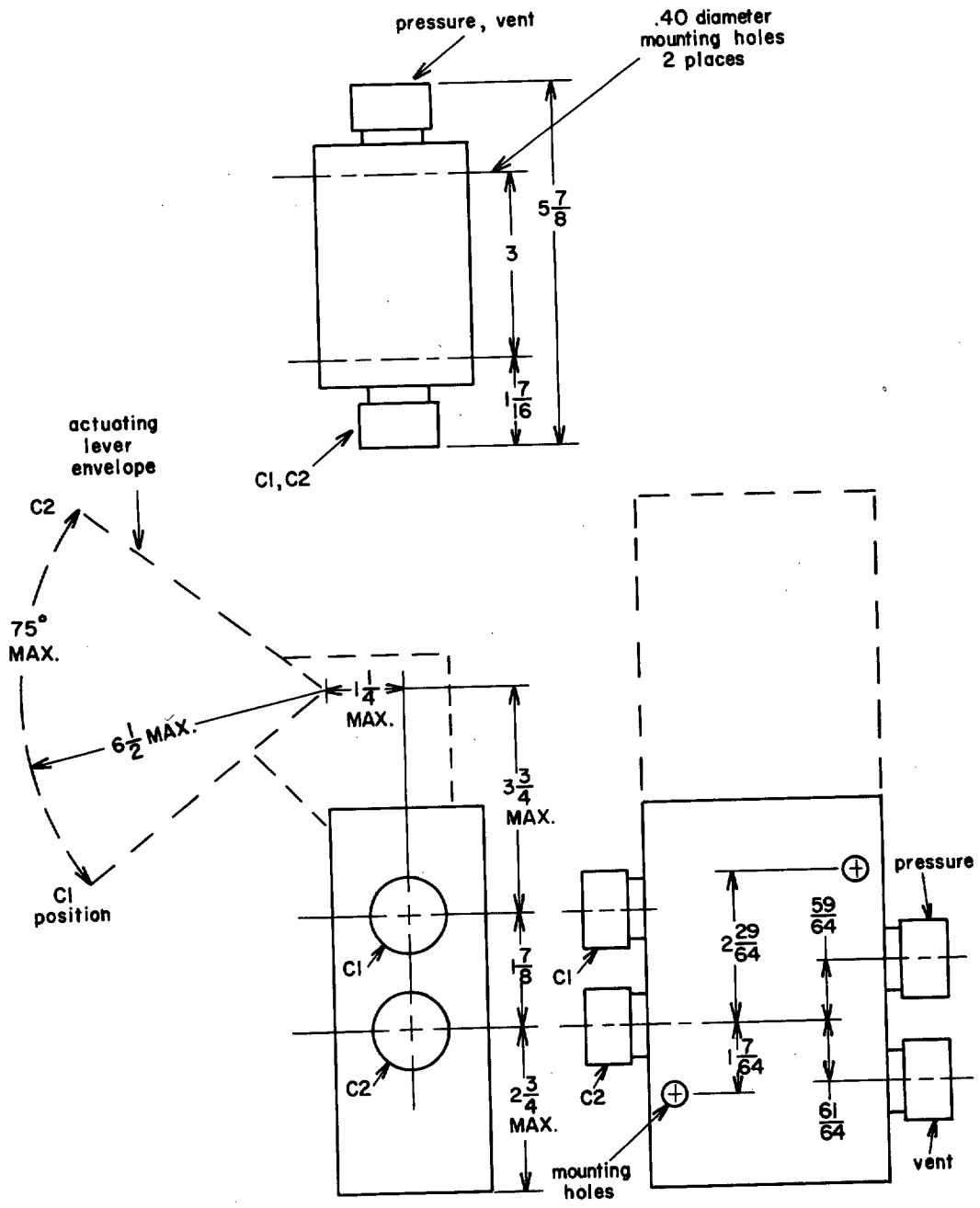


Figure 1 - Pattern A (24760-1).

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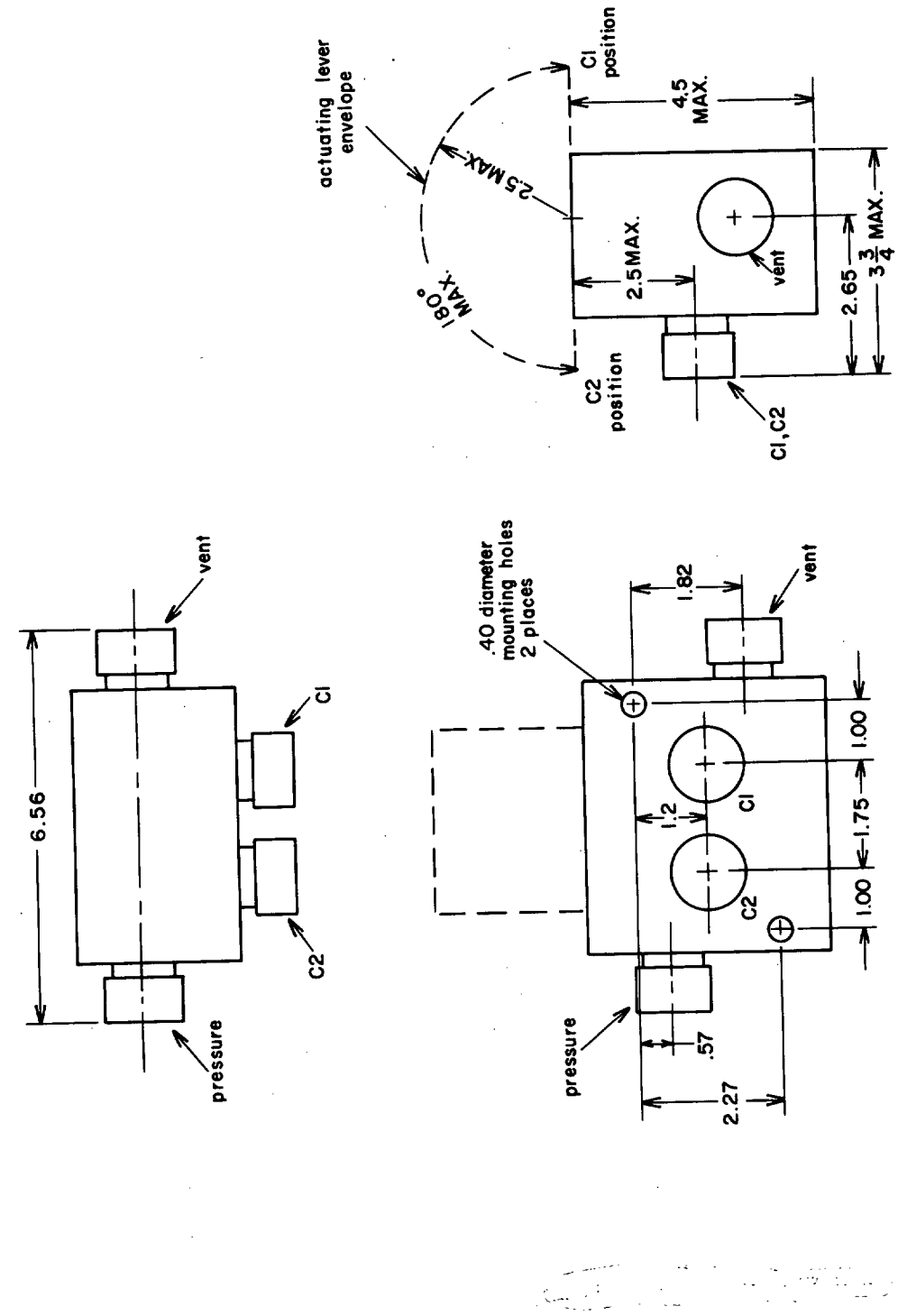


Figure 2 - Pattern B (LV-15A).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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